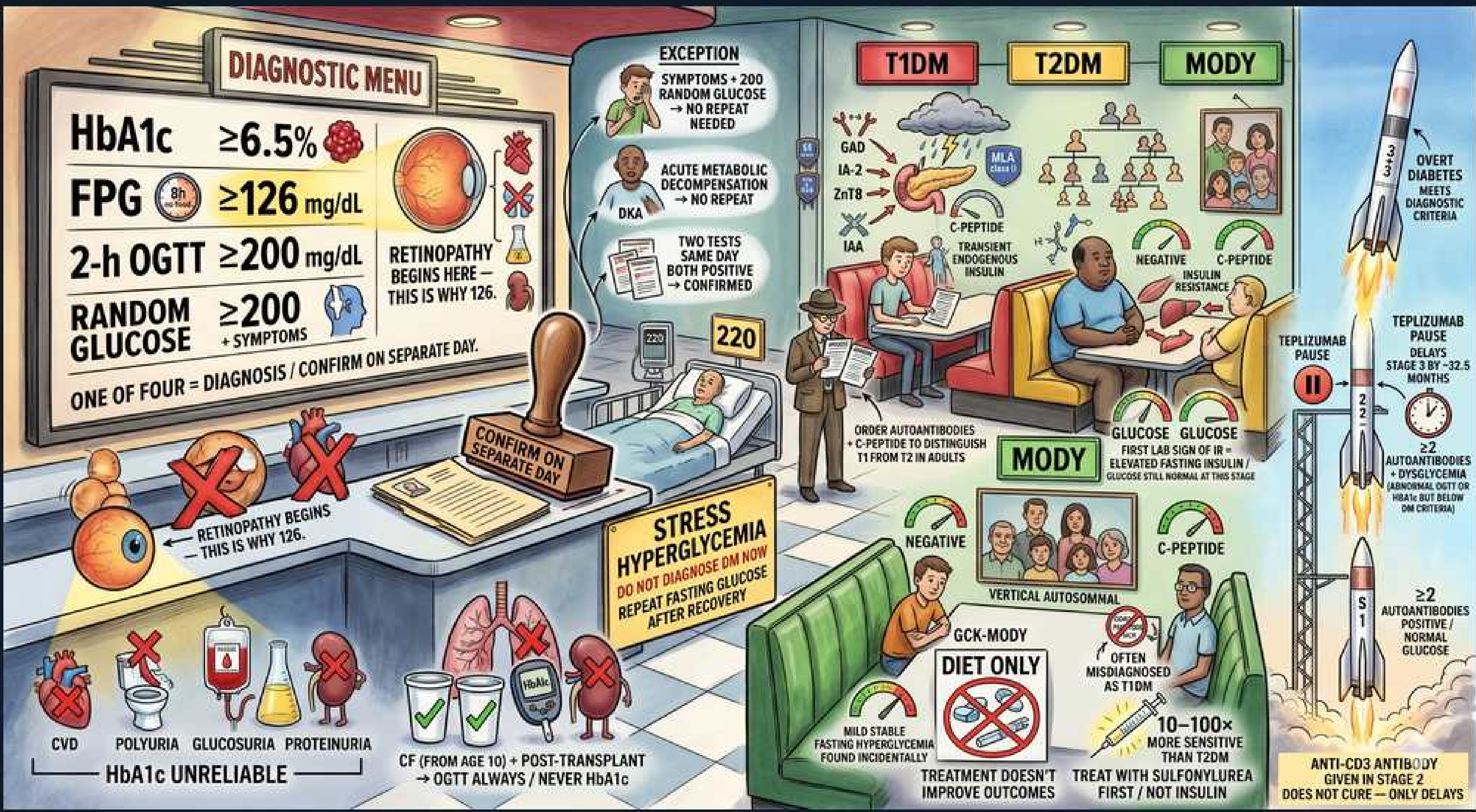
Diabetes — Dx Overview 1: Diagnostic Menu & Classification



1. Diagnostic Menu (4 criteria)

WHAT YOU SEE

Red 'Diagnostic Menu' board: HbA1c >=6.5%, FPG >=126 mg/dL, 2-h OGTT >=200 mg/dL, Random glucose >=200 + symptoms.

WHAT IT ENCODES

Diabetes is diagnosed by ANY ONE of four criteria: (1) HbA1c >=6.5%; (2) fasting plasma glucose (FPG) >=126 mg/dL (fasting = no calories >=8 h); (3) 2-hour glucose >=200 mg/dL on a 75 g OGTT; (4) random plasma glucose >=200 mg/dL WITH classic hyperglycemic symptoms. Confirm with a repeat test on a separate day unless unequivocal hyperglycemia. Mnemonic: 'A1c 6.5, Fasting 126, OGTT/Random 200.'

BOARD TRAP

Watch the exact cutoffs: FPG is 126 (NOT 120 or 140), A1c is 6.5% (NOT 7% — 7% is a treatment target, not a diagnostic threshold), and a random glucose >=200 only counts WITH symptoms (asymptomatic random glucose is not diagnostic).

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2. Confirm on separate day

WHAT YOU SEE

Stamp: 'Confirm on separate day.'

WHAT IT ENCODES

A single abnormal test in an ASYMPTOMATIC patient must be CONFIRMED — ideally by repeating the same test, on a separate day. If two different tests (e.g., A1c and FPG) are both above threshold, that also confirms the diagnosis without repeating.

BOARD TRAP

EXCEPTION: a patient with unequivocal hyperglycemia plus classic symptoms or acute metabolic decompensation (DKA/HHS) needs NO confirmatory repeat — the trap is demanding a second test before treating an obviously symptomatic/decompensated patient.

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3. Exception — no repeat needed

WHAT YOU SEE

'Exception' box: symptoms + random glucose >200, or acute metabolic decompensation (DKA).

WHAT IT ENCODES

Diagnose immediately without a second test when there are classic symptoms (polyuria, polydipsia, weight loss) plus a random plasma glucose >=200 mg/dL, OR acute metabolic decompensation such as DKA or HHS. These are 'unequivocal hyperglycemia.'

BOARD TRAP

Two DIFFERENT tests positive at the SAME draw (e.g., A1c >=6.5% and FPG >=126 on one sample) also confirms the diagnosis — the trap is insisting confirmation must always be on a separate day; concordant same-day tests suffice.

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4. Retinopathy = why 126

WHAT YOU SEE

Eye/retina image; 'Retinopathy begins here — this is why 126.'

WHAT IT ENCODES

The FPG threshold of 126 mg/dL (and A1c 6.5%) was chosen because the prevalence of diabetic RETINOPATHY rises sharply above these values — the cutoffs are anchored to microvascular complication risk, not arbitrary statistics.

BOARD TRAP

Don't assume the diagnostic cutoffs are based on cardiovascular risk — they are complication (retinopathy)-anchored; CV risk rises continuously even within the 'prediabetic' range.

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5. HbA1c unreliable settings

WHAT YOU SEE

Bottom-left red X's: CVD, polyuria, glucosuria, proteinuria, hemoglobinopathies, transfusion, pregnancy.

WHAT IT ENCODES

A1c reflects average glycemia over ~3 months (RBC lifespan ~120 days), so it is UNRELIABLE whenever RBC turnover or hemoglobin is altered: hemolytic anemia, recent blood loss/transfusion, iron/B12 deficiency anemia, pregnancy, advanced CKD/ESRD (and erythropoietin use), and hemoglobinopathies (HbS, HbC, thalassemia). Conditions that SHORTEN RBC lifespan falsely LOWER A1c; longer survival falsely RAISES it.

BOARD TRAP

In any altered-RBC-turnover state, do NOT use A1c — switch to FPG or OGTT. The trap is diagnosing or excluding diabetes by A1c in a patient with hemolysis, recent transfusion, pregnancy, or a hemoglobinopathy.

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6. Stress hyperglycemia

WHAT YOU SEE

'Stress hyperglycemia — do NOT diagnose in hospital; repeat fasting glucose after recovery.'

WHAT IT ENCODES

Acute illness, infection, surgery, MI, and stress (cortisol/catecholamines, exogenous glucocorticoids) transiently raise glucose. Do NOT diagnose diabetes during acute illness — an A1c can help retrospectively (>=6.5% suggests pre-existing diabetes), but confirm with FPG/OGTT after recovery.

BOARD TRAP

A high glucose in an acutely ill inpatient is NOT diagnostic of diabetes — the trap is labeling a sick, stressed patient diabetic on the basis of inpatient hyperglycemia alone; recheck after recovery (an A1c is the way to detect pre-existing disease).

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7. CF & post-transplant — OGTT

WHAT YOU SEE

CF (from age 10): OGTT always, never A1c; post-transplant -> OGTT.

WHAT IT ENCODES

Cystic fibrosis-related diabetes (CFRD) is screened annually from age 10 with the OGTT (NEVER A1c — altered RBC turnover and the OGTT's higher sensitivity in CF). Post-transplant diabetes mellitus (NODAT/PTDM) is likewise diagnosed with OGTT once the patient is stable on maintenance immunosuppression (tacrolimus, steroids are diabetogenic).

BOARD TRAP

Using A1c to screen CF or post-transplant patients is the trap — A1c underestimates glycemia in both; OGTT is the standard. Also screen for PTDM only after the patient is stable, not during the high-steroid early post-op period.

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8. T1DM panel

WHAT YOU SEE

'T1DM' arch: storm cloud (autoimmune), GAD65, ZnT8, IA-2, IAA antibodies; low C-peptide.

WHAT IT ENCODES

Type 1 DM = autoimmune T-cell-mediated beta-cell destruction → absolute insulin deficiency. Islet autoantibodies: GAD65, IA-2 (ICA512), insulin (IAA), and ZnT8. C-peptide is LOW/undetectable (endogenous insulin gone). Prone to DKA; requires insulin. Associated with other autoimmune disease (thyroid, celiac).

BOARD TRAP

A single autoantibody is not strongly diagnostic — multiple antibodies confer higher risk. GAD65 is the most persistent in adults and defines LADA (latent autoimmune diabetes of adults), an adult-onset T1 variant often MISLABELED as T2DM; suspect LADA when a 'lean T2DM' fails oral agents quickly. C-peptide distinguishes T1 (low) from T2 (normal/high).

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9. T2DM panel

WHAT YOU SEE

'T2DM' arch: couch potatoes, insulin resistance, transient endogenous insulin, normal/high C-peptide.

WHAT IT ENCODES

Type 2 DM = insulin RESISTANCE plus a relative (progressive) insulin secretory defect, driven by obesity/inactivity/genetics. C-peptide is normal or ELEVATED early (hyperinsulinemia), declining over years. Onset is gradual and often asymptomatic; autoantibodies negative. Strong family history; associated with metabolic syndrome and acanthosis nigricans.

BOARD TRAP

Don't assume every adult with hyperglycemia has T2DM — a lean adult who is antibody-positive with low C-peptide and rapid oral-agent failure is LADA/T1, not T2DM. Conversely, obese adolescents can have T2DM (rising incidence).

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10. MODY

WHAT YOU SEE

'MODY' arch: rocket (autosomal dominant), GCK-MODY = mild stable fasting hyperglycemia.

WHAT IT ENCODES

MODY (maturity-onset diabetes of the young) = monogenic, AUTOSOMAL DOMINANT diabetes presenting in young (<25), typically non-obese patients with a strong multigenerational (vertical) family history. Autoantibodies NEGATIVE, C-peptide PRESERVED. Common subtypes: GCK (glucokinase) and HNF1A/HNF4A.

BOARD TRAP

MODY is NOT autoimmune (antibodies negative) and NOT typical T2DM — the discriminator is autosomal dominant VERTICAL inheritance across generations with young, lean onset and preserved C-peptide. The trap is calling a thin antibody-negative young diabetic 'early T2DM' and missing MODY (genetic testing changes treatment).

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11. MODY treatment / GCK

WHAT YOU SEE

GCK-MODY: diet only, treatment doesn't improve outcomes; other MODY: treat with sulfonylurea first, not insulin.

WHAT IT ENCODES

GCK-MODY (MODY2) = mild, stable, lifelong fasting hyperglycemia (~100-150 mg/dL) that is BENIGN and needs DIET ONLY — pharmacotherapy does not improve outcomes (except sometimes in pregnancy). HNF1A-MODY (MODY3) and HNF4A-MODY (MODY1) are exquisitely SULFONYLUREA-sensitive — treat with low-dose sulfonylurea first, not insulin.

BOARD TRAP

MODY responds to SULFONYLUREAS — the trap is defaulting these young patients to insulin (as if T1DM); HNF1A/HNF4A MODY are sulfonylurea-responsive, and GCK-MODY usually needs no drug at all.

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12. Teplizumab — delay T1DM

WHAT YOU SEE

'Teplizumab pause' rocket: Stage 2 (>=2 autoantibodies + dysglycemia, normal glucose) delays Stage 3.

WHAT IT ENCODES

Teplizumab (anti-CD3 monoclonal) is given in STAGE 2 T1DM (>=2 islet autoantibodies + dysglycemia but NOT yet overt diabetes) to DELAY progression to clinical Stage 3 disease by a median of ~2 years. Staging: Stage 1 = >=2 antibodies, normoglycemia; Stage 2 = >=2 antibodies + dysglycemia; Stage 3 = clinical hyperglycemia/symptoms.

BOARD TRAP

Teplizumab only DELAYS onset — it does not prevent or cure T1DM, and it is used in Stage 2 (presymptomatic dysglycemia), not in established/Stage 3 disease. The trap is expecting cure or using it after clinical diabetes has developed.
